AI + Quantum AI Senior Engineer Learning Plan (v2)

Senior-Level Full-Stack AI + Quantum AI — Learning Plan (v2, time-budgeted)

Owner: MD Hashim · **Window:** 13 Jul 2026 → 31 Jan 2027 (~29 weeks) · **Load:** 33–35 hrs/week (weekend-loaded; see template) · **Total:** ~975 hrs

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What changed from v1 (read this first)

You asked for three things, and they reshaped the plan:

1. **"Best standard time" per topic.** Every block below now carries a realistic hour estimate — the time the material actually takes when done properly (watch → re-derive → implement → read the paper), not a compressed guess. These add up honestly; I've budgeted them against the calendar.
2. **Finish by end of Jan 2027.** The window is now ~29 weeks. Because that's tighter than v1's timeline but you allowed more weekly hours, the plan runs at **33–35 hrs/week with the heavy lifting on weekends** rather than v1's flat 30.
3. **Prioritize by role need + prerequisites.** The phase order changed. See the logic below.

Prioritization & sequencing logic

Two forces decide the order: **what your job needs first**, and **what has to be learned before what**.

- **Linear algebra is the universal prerequisite** — it sits under deep learning *and* quantum. So it is hard-front-loaded into Weeks 1–2, before anything else, and continues as a thinner parallel thread after.
- **Brain segmentation is your active project → Phase 1.** It's the immediate fire. Its own sub-prerequisites (convolution arithmetic, enough probability for loss functions) are pulled in right where they're needed.
- **Quantum ML is your August project → Phase 2 (moved up from v1's Phase 3).** A potential project in August can't sit behind a general deep-learning deep-dive. Phase 2's first two weeks are a **hands-on readiness sprint** (practical, tooling-first) so you're productive the moment the project lands — leaning on the VQE/ADAPT-VQE/DMET-VQE experience you already have — and the remaining weeks build the senior-level *theory* depth underneath that experience.
- **Modern DL theory (transformers/generative) → Phase 3.** This is *deepening* skills you already use in production, and it's a prerequisite for the full-stack/LLM phase — so it comes after the two role-critical phases but before production.
- **Full-stack / production → Phase 4**, then **specialization + integration + interview → Phase 5.** Valuable, but least time-critical relative to your immediate role.

> **If the August quantum project starts before you finish Phase 1:** pull the Phase 2 readiness sprint (Weeks 8–9) forward and run it in parallel with your remaining Phase 1 weeks. Your existing hands-on quantum background makes this dual-track feasible for a 2–3 week overlap. Don't parallelize the *theory* weeks — those need undivided focus.

Phase summary (the whole plan at a glance)

Phase	Focus	Weeks	Approx. dates	Phase hrs
1	Prerequisite math sprint + **Medical Image Segmentation** (active project)	1–7	13 Jul → 30 Aug	~235
2	**Quantum Computing + QML** (August project) — readiness sprint then theory depth	8–16	31 Aug → 1 Nov	~265
3	Modern Deep Learning theory + foundation architectures	17–21	2 Nov → 6 Dec	~150
4	Full-stack / production AI systems	22–26	7 Dec → 10 Jan	~160
5	Specialization + integration + interview prep	27–29	11 Jan → 31 Jan	~110

Track M (mathematics) runs *underneath* all phases; its hours are folded into the phase totals above.

Weekly time template (33–35 hrs, weekend-loaded)

	Deep hands-on / capstone	Lectures + paper reading	Math (Track M)	Whiteboard + writeup
Mon–Fri (~3 hrs/day = 15)	3	8	3	1
Sat–Sun (~9–10 hrs/day = 18–20)	11	5	3	1
Weekly total	**14**	**13**	**6**	**2**

Weekends carry the cognitively heavy work — reproducing nnU-Net, the quantum capstone, re-deriving proofs — because that's when you have uninterrupted blocks. Weekdays are for lectures and reading you can do in shorter sittings. **Dial down to ~30 on brutal work weeks; push to ~40 on light ones.** The plan assumes you'll flex, not that every week is identical.

One honest caution: 33–35 hrs of learning *on top of a full-time senior engineering role* is a lot, sustained over seven months. It's doable because you allowed weekend loading and because much of this deepens work you already do — but protect at least one genuine rest block per week. Burnout in month three costs more than the hours it "saves."

PHASE 1 — Prerequisite Math Sprint + Medical Image Segmentation

Weeks 1–7 (13 Jul → 30 Aug 2026) · ~235 hrs · PRIORITY: immediate (active project)

Goal: Own MRI / fluorescence brain segmentation end-to-end — physics → preprocessing → architectures → losses → 3D → uncertainty → evaluation — while front-loading the linear algebra that unblocks everything after. By end of phase you can whiteboard nnU-Net's design choices cold and defend each one in a project discussion.

Weeks 1–2 — Linear algebra hard-front-load + imaging physics (~70 hrs)

Do these two in parallel: math in the weekday slots, physics on the weekends. This is the prerequisite burst.

Linear algebra (~40 hrs):

- **3B1B — Essence of Linear Algebra** (build the geometric intuition first; watch twice if needed) — *~8 hrs* https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
- **MIT 18.06 — Gilbert Strang** (selected: L1, 3, 5, 6, 9, 10, 14, 15, 16, 21, 22, 25, 29, 30 + problems) — *~30 hrs* https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/
- **Reference kept open:** Matrix Cookbook — https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf

Imaging physics (~30 hrs) — most engineers skip this and it shows in their model design:

- **Allen Elster — Questions & Answers in MRI** (browse; the reference you'll return to) — *~8 hrs* https://mriquestions.com/index.html
- **OHBM Educational — MRI physics** playlist — *~8 hrs* https://www.youtube.com/@ohbmonline/playlists
- **Paul Callaghan — "Magnetic Resonance" (Otago)**, first 4 lectures — *~6 hrs* https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z
- **iBiology — fluorescence microscopy** primer (your tissue work needs this) — *~8 hrs* https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMl3PlpdW9TMP

Deliverable: 1-page note — T1 vs T2 vs FLAIR vs T1CE (what tissue looks like in each, why each exists, what artifacts you must preprocess: motion, bias field, partial volume); for fluorescence: photobleaching, channel bleed-through, autofluorescence and their effect on segmentation. **Whiteboard checkpoint:** derive SVD from eigendecomposition; explain geometrically what a rank-r approximation does; show why A^TA is PSD; derive backprop through a linear layer with matrix calculus.

Week 3 — CNNs, U-Net family, convolution math (~35 hrs)

- **Stanford CS231n 2017 — Justin Johnson** (focus L5, 9, 11 + notes) — *~14 hrs* Playlist: https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv · Notes: https://cs231n.github.io/
- **Convolution arithmetic (Dumoulin & Visin)** — the canonical reference for shapes/strides/padding/dilation in 2D *and* 3D — *~5 hrs* https://arxiv.org/abs/1603.07285 · Animations: https://github.com/vdumoulin/conv_arithmetic
- **U-Net family papers, read in order, derive output sizes by hand** — *~16 hrs*
  - U-Net (Ronneberger 2015): https://arxiv.org/abs/1505.04597
  - V-Net (3D): https://arxiv.org/abs/1606.04797
  - Attention U-Net: https://arxiv.org/abs/1804.03999
  - **nnU-Net (most important for you — read twice + supplementary):** https://www.nature.com/articles/s41592-020-01008-z

Whiteboard checkpoint: draw U-Net from memory; compute output size at every layer for 256×256 input; explain *why* skip connections work; explain why nnU-Net's "no new architecture" thesis is profound.

Week 4 — Loss functions for medical segmentation (your 95/5 imbalance) (~30 hrs)

This is where segmentation pipelines die. Own it cold.

- **Survey of segmentation loss functions (Jadon)** — derive each — *~6 hrs* https://arxiv.org/abs/2006.14822
- **Generalized Dice (Sudre 2017)** — https://arxiv.org/abs/1707.03237 — *~3 hrs*
- **Focal Tversky (severe imbalance — your friend)** — https://arxiv.org/abs/1810.07842 — *~3 hrs*
- **Boundary Loss (thin structures)** — https://arxiv.org/abs/1812.07032 — *~3 hrs*
- **From-scratch PyTorch implementation deliverable** — Dice, GDL, Focal, Tversky, Focal Tversky, Boundary, compound DiceCE; unit-test on synthetic imbalanced data; plot gradients vs class ratio; document when each helps — *~15 hrs*

Week 5 — nnU-Net deep dive + MONAI ecosystem + DigitalSreeni (~45 hrs)

The biggest hands-on block in Phase 1. nnU-Net stops being a black box here.

- **MONAI tutorials** (clone, run, modify the brain-seg notebooks) — *~15 hrs* https://github.com/Project-MONAI/tutorials · Videos: https://www.youtube.com/@projectmonai/videos
- **nnU-Net v2** — read the *source*, it teaches more than the paper — *~8 hrs* https://github.com/MIC-DKFZ/nnUNet · MONAI integration: https://github.com/Project-MONAI/tutorials/blob/main/nnunet/README.md · Kitware comparison: https://www.kitware.com/developing-custom-3d-medical-image-segmentation-solutions-using-out-of-the-box-pipelines-in-monai/
- **DigitalSreeni (Python for Microscopists)** — the most microscopy-native channel there is; far closer to your fluorescence + volumetric work than any Stanford course. Pick ~6–8 videos mapped to your bottlenecks; do them code-along — *~14 hrs* Channel: https://www.youtube.com/@DigitalSreeni/playlists · Code: https://github.com/bnsreenu/python_for_microscopists · U-Net playlist: https://www.youtube.com/playlist?list=PLZsOBAyNTZwbR08R959iCvYT3qzhxvGOE
- **AI for Medical Diagnosis (Rajpurkar, deeplearning.ai)** — audit free; the brain-MRI U-Net segmentation module + the evaluation module are on-point. Skip the X-ray module if pressed — *~8 hrs* https://www.coursera.org/learn/ai-for-medical-diagnosis

Deliverable: reproduce nnU-Net on a public brain dataset (BraTS subset or a Medical Segmentation Decathlon task).

Write up which defaults it picked, *why*, and what you'd change for fluorescence microscopy where statistics differ from MRI.

Week 6 — Transformer-based segmentation + foundation models (~25 hrs)

Where the field is now, and where senior interview questions are heading.

- **ViT** (re-derive attention math) — https://arxiv.org/abs/2010.11929 — *~4 hrs*
- **TransUNet / Swin-UNet / UNETR / Swin UNETR** — *~10 hrs* https://arxiv.org/abs/2102.04306 · https://arxiv.org/abs/2105.05537 · https://arxiv.org/abs/2103.10504 · https://arxiv.org/abs/2201.01266
- **SAM / MedSAM** — https://arxiv.org/abs/2304.02643 · https://www.nature.com/articles/s41467-024-44824-z — *~6 hrs*
- **MedNeXt** (a ConvNet that beats transformers on many medical tasks — keeps you honest) — https://arxiv.org/abs/2303.09975 — *~5 hrs*

Whiteboard checkpoint: compare inductive biases of CNN vs ViT vs hybrid — when does each win? Role of foundation models in low-data medical regimes? Walk through SAM's promptable design and what MedSAM changes.

Week 7 — 3D segmentation, uncertainty, evaluation + capstone (~55 hrs)

- **3D pipelines with TorchIO** (augmentation done right for volumes) — *~12 hrs* https://torchio.readthedocs.io/ · https://www.youtube.com/playlist?list=PLfXIfDgnRPlflEHFx2N6mlbn-MGFOPiX2
- **Uncertainty (Yarin Gal — MC Dropout, deep ensembles, evidential)** — *~8 hrs* https://www.youtube.com/results?search_query=yarin+gal+uncertainty+deep+learning
- **Metrics Reloaded** (the definitive 2024 paper on choosing eval metrics — must-read) — *~5 hrs* https://www.nature.com/articles/s41592-023-02151-z
- **Domain shift in medical imaging (Stanford AIMI)** — *~5 hrs* https://aimi.stanford.edu/education/educational-resources
- **3D-conv whiteboard exercise:** for input (D=64, H=128, W=128, C=4), kernel (3,3,3), stride (1,2,2), padding (1,1,1), 32 output channels — compute output shape AND param count by hand; repeat for transposed and dilated conv — *~3 hrs*
- **PHASE 1 CAPSTONE** *(~22 hrs)* — self-contained notebook + 3-page write-up on your mouse-brain data (or a BraTS subset): **(1)** preprocessing decisions with physics justification, **(2)** architecture ablation across 3 architectures, **(3)** loss-function ablation, **(4)** uncertainty quantification + failure-mode analysis. This is the artifact a senior engineer produces and exactly what a panel wants to see.

Phase 1 interview self-test: whiteboard U-Net forward pass + skip connections with shapes; derive Dice gradient and explain instability on empty masks; explain why nnU-Net usually beats fancier architectures (and when it doesn't); explain attention mathematically + ViT patch embedding; for a new dataset, walk through diagnosing bias-field artifacts, class imbalance, label noise, domain shift.

PHASE 2 — Quantum Computing + Quantum Machine Learning

Weeks 8–16 (31 Aug → 1 Nov 2026) · ~265 hrs · PRIORITY: high (August project)

Goal: Match your existing hands-on VQE/ADAPT-VQE/DMET-VQE work with rigorous theory. Derive VQE from the variational principle, explain barren plateaus mathematically, defend ansatz choices, and discuss QML's real limits honestly (per Schuld's recent critiques). This is *depth on top of experience*, not from-zero.

Weeks 8–9 — HANDS-ON READINESS SPRINT (~55 hrs)

Front-loaded and practical so you're productive the instant the August project lands. *(This is the block to pull forward / parallelize with late Phase 1 if the project starts early.)*

- **PennyLane Codebook — Variational Quantum Algorithms** (work the exercises) — *~15 hrs* https://pennylane.ai/codebook
- **PennyLane quantum-chemistry demos** — you've done this in practice; now run *all* the derivations behind the code — *~15 hrs* https://pennylane.ai/qml/demos_quantum-chemistry
- **Qiskit Global Summer School — recent recordings** (chemistry + variational tracks) — *~12 hrs* https://www.youtube.com/@qiskit/playlists

- **IBM Quantum Learning platform** — set up, run current tooling end-to-end — *~8 hrs* https://learning.quantum.ibm.com/
- **Sample-based / Krylov Quantum Diagonalization (QDA)** — IBM's newest near-term technique — *~5 hrs* https://www.ibm.com/quantum/blog/iql-migration

Week 10 — Quantum-mechanics math (Hilbert spaces, operators, tensor products) (~25 hrs)

The prerequisite for the theory weeks. Do it alongside the readiness sprint's tail if needed.

- **MIT 8.04 — Quantum Physics I (Adams)** (L6–12 if the experimental motivation feels slow) — *~15 hrs* https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/video_galleries/lecture-videos/
- **Quantum Country** (spaced-repetition intro — fast and sticky) — *~6 hrs* https://quantum.country/qcvc
- **(Optional, for the rigor-minded)** Frederic Schuller — Geometric Anatomy of Theoretical Physics — *~4 hrs sampled* https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic

Whiteboard checkpoint: define a Hilbert space rigorously; tensor-product two qubit states; explain unitarity, Hermitian operators, why measurement outcomes are eigenvalues; distinguish pure vs mixed states + the density matrix.

Weeks 11–13 — Quantum computing fundamentals: the Watrous / IBM track (~80 hrs)

The best free quantum-information series that exists. John Watrous's four IBM courses.

- **Basics of Quantum Information** — *~25 hrs* https://learning.quantum.ibm.com/course/basics-of-quantum-information · YouTube: https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR
- **Fundamentals of Quantum Algorithms** — *~25 hrs* https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms
- **General Formulation of Quantum Information** (density matrices, channels, POVMs — matters for noise/error mitigation) — *~18 hrs* https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information
- **Foundations of Quantum Error Correction** — *~12 hrs* https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction
- **Companion (free):** Nielsen & Chuang, *Quantum Computation and Quantum Information*, ch 1–4, 8, 10 (read alongside)

Weeks 14–15 — Variational algorithms + QML theory (~65 hrs)

You've shipped this — now own the math and the critical literature.

Variational (~35 hrs):

- Variational principle + VQE end-to-end, re-derived — Peruzzo 2014: https://www.nature.com/articles/ncomms5213
- ADAPT-VQE (why it's parameter-efficient) — Grimsley 2019: https://www.nature.com/articles/s41467-019-10988-2
- DMET-VQE / fragment methods (your stack) — survey: https://arxiv.org/abs/2108.08987
- QAOA — Farhi 2014: https://arxiv.org/abs/1411.4028 · Hadfield review: https://arxiv.org/abs/1709.03489
- Pauli decomposition + measurement grouping (shot-budget engineering): https://arxiv.org/abs/1907.13117

QML (~30 hrs):

- **PennyLane QML topic page (Schuld-curated)** — the most honest overview in the field: https://pennylane.ai/topics/quantum-machine-learning
- **Maria Schuld — "Taking stock of QML: a critical perspective"** — the talk every senior should watch: https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning
- Schuld & Petruccione, *Machine Learning with Quantum Computers* — ch 4–7 (kernels, variational, QNNs)
- Key papers: "QML models are kernel methods" (https://arxiv.org/abs/2101.11020) · Barren plateaus (https://www.nature.com/articles/s41467-018-07090-4) · Expressibility & entangling capability (https://arxiv.org/abs/1905.10876) · "Is quantum advantage the right goal?" (https://arxiv.org/abs/2203.01340) · PQC encodings (https://arxiv.org/abs/2008.08605)
- Structured courses to pull lectures from: Christophe Péré (https://github.com/Christophe-pere/QML-Course) · Morten Hjorth-Jensen, Oslo (https://github.com/CompPhysics/QuantumComputingMachineLearning)

Whiteboard checkpoint: derive the variational principle; walk VQE end-to-end (ansatz choice, parameter-shift rule, optimizer); explain barren plateaus mathematically + mitigations; why is ADAPT-VQE more parameter-efficient; when is DMET fragmentation worth the overhead.

Week 16 — Quantum-chemistry capstone (~40 hrs)

PHASE 2 CAPSTONE: take one molecule from your Lilly work (keto-enol tautomerism or HATU coupling). Re-implement three ways — vanilla VQE + UCCSD, ADAPT-VQE, DMET-VQE — benchmark against classical CCSD/6-31G. The 4-page write-up defends *every* choice on theoretical grounds (ansatz family, optimizer, measurement grouping, noise mitigation, basis set) and includes a barren-plateau analysis. Exactly what a senior QML interview wants.

Phase 2 interview self-test: derive VQE from the variational principle; explain the parameter-shift rule and why it works; whiteboard a barren-plateau argument; VQE vs QAOA — when each, and when *no* quantum algorithm; honest take on where QML is overhyped vs promising (cite Schuld); walk through DMET fragmentation and the classical-quantum interface.

PHASE 3 — Modern Deep Learning Theory + Foundation Architectures

Weeks 17–21 (2 Nov → 6 Dec 2026) · ~150 hrs · PRIORITY: medium (deepening + prereq for Phase 4)

Goal: Stop treating transformers as a black box. Implement GPT from scratch, derive every layer of attention, understand modern training tricks, and defend architecture choices on theoretical grounds.

Weeks 17–18 — Karpathy "Neural Networks: Zero to Hero" (~55 hrs)

Non-negotiable even with your experience. Do all 10 videos *with the exercises* — re-implement each without looking at his code. That's what makes the difference.

- Playlist: https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ · Hub: https://karpathy.ai/zero-to-hero.html
- Order: micrograd → makemore 1–5 (incl. BatchNorm internals, manual backprop, WaveNet) → GPT from scratch → tokenizer → reproduce GPT-2.

Week 19 — Transformers in depth + modern variants (~40 hrs)

- **Stanford CS25 — Transformers United** (pick 6–8 talks) — https://web.stanford.edu/class/cs25/ · https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM
- **Lilian Weng's blog** (attention, transformer, RLHF posts) — https://lilianweng.github.io/
- **The Annotated Transformer** (code + math line by line) — http://nlp.seas.harvard.edu/annotated-transformer/
- **Modern tricks** (~30 min each): RoPE (https://arxiv.org/abs/2104.09864) · RMSNorm (https://arxiv.org/abs/1910.07467) · FlashAttention (https://arxiv.org/abs/2205.14135) · GQA/MQA (https://arxiv.org/abs/2305.13245) · SwiGLU (https://arxiv.org/abs/2002.05202) · MoE / Switch (https://arxiv.org/abs/2101.03961)

Whiteboard checkpoint: derive scaled dot-product attention; explain the $\sqrt{d_k}$ scaling; multi-head as a single matmul; RoPE geometrically; why pre-LN beats post-LN.

Week 20 — Vision transformers + ConvNets in the modern era (~22 hrs)

- ViT lectures (Lucas Beyer) — https://www.youtube.com/results?search_query=lucas+beyer+vision+transformer
- ConvNeXt (https://arxiv.org/abs/2201.03545) · DINOv2 (self-supervised, relevant to label-scarce medical work — https://arxiv.org/abs/2304.07193) · MAE (https://arxiv.org/abs/2111.06377)
- **Hugging Face Computer Vision course** — https://huggingface.co/learn/computer-vision-course

Week 21 — Diffusion + generative modeling theory (~33 hrs)

Diffusion math underpins the modern field and shows up in medical image synthesis/augmentation.

- **Stanford CS236 — Deep Generative Models (Ermon)** (L1–3, 6–8, 11–13) — https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8
- DDPM (https://arxiv.org/abs/2006.11239) · Yang Song, score-based unification + blog (https://yang-song.net/blog/2021/score/) · Lilian Weng diffusion post (https://lilianweng.github.io/posts/2021-07-11-diffusion-models/)
- **Hugging Face Diffusion course** (units 1–2 hands-on) — https://huggingface.co/learn/diffusion-course

Phase 3 interview self-test: implement attention on a whiteboard in 5 min; derive the diffusion loss (variational + score-matching views); why LayerNorm not BatchNorm in transformers; positional encodings compared (sinusoidal/learned/RoPE/ALiBi); compute a transformer block's param count from d_model, n_heads, FFN ratio.

PHASE 4 — Full-Stack / Production AI Systems

Weeks 22–26 (7 Dec 2026 → 10 Jan 2027) · ~160 hrs · PRIORITY: medium (deepening ongoing work)

Goal: Upgrade from "I built it" to "I can defend every architectural choice and design the next-gen version" across LLM systems, RAG, MLOps, distributed training, and serving.

Week 22 — LLM internals (~30 hrs)

- **Stanford CS336 — Language Modeling from Scratch** (currently the best LLM-systems course) — https://stanford-cs336.github.io/spring2024/ · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_
- **Hugging Face NLP course** (units new to you) — https://huggingface.co/learn/nlp-course

Week 23 — RAG at senior level (~28 hrs)

- **Hamel Husain — "What we learned from a year of building with LLMs"** (Parts I–II) — https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/
- **Eugene Yan — LLM patterns** — https://eugeneyan.com/writing/llm-patterns/
- **Anthropic — contextual retrieval** — https://www.anthropic.com/news/contextual-retrieval
- Reranking, hybrid search, query rewriting, evals; **LlamaIndex** (https://docs.llamaindex.ai/) · **RAGAS** (https://docs.ragas.io/) · **DSPy** (https://dspy.ai/)

Week 24 — MLOps + ML systems design (~42 hrs)

- **Chip Huyen — *Designing Machine Learning Systems*** / interviews book — https://huyenchip.com/ml-interviews-book/contents/8.1.1-ml-systems-design.html
- **Stanford CS329S — ML Systems Design** — https://stanford-cs329s.github.io/
- **Made With ML** (hands-on) — https://madewithml.com/ · **Full Stack Deep Learning** — https://fullstackdeeplearning.com/course/2022/

Week 25 — Distributed training + inference optimization (~25 hrs)

- **Sebastian Raschka** (distributed training + efficient fine-tuning) — https://magazine.sebastianraschka.com/
- **HF — efficient training on multiple GPUs** — https://huggingface.co/docs/transformers/en/perf_train_gpu_many
- **vLLM** (paper + run it) — https://arxiv.org/abs/2309.06180 · quantization (GPTQ/AWQ/GGUF/FP8) · **ZeRO** — https://arxiv.org/abs/1910.02054

Week 26 — Eval / observability / safety + systems-design interview prep (~35 hrs)

- **Anthropic — Constitutional AI** — https://www.anthropic.com/news/constitutional-ai · **Eugene Yan — LLM evals** — https://eugeneyan.com/writing/llm-evaluators/
- Observability tool (LangSmith / Phoenix / Helicone) — run one on a real RAG system.
- **Systems-design interview prep** — Chip Huyen's book (https://huyenchip.com/ml-interviews-book/); practice prompts on a whiteboard, recorded, self-critiqued: "RAG assistant for HCPs, 10K queries/day on 50K docs"; "brain-seg pipeline ingesting new-institution data weekly with domain-shift handling"; "retraining + monitoring for a drug-discovery digital twin."

PHASE 5 — Specialization + Integration + Interview Prep

Weeks 27–29 (11 Jan → 31 Jan 2027) · ~110 hrs · PRIORITY: capstone

Goal: Close the geometric-deep-learning / molecular-AI gap that ties your imaging, quantum, and drug-discovery work together — then integrate everything and run mock interviews.

Week 27 — Geometric DL + molecular ML (~40 hrs)

- **Bronstein et al. — Geometric Deep Learning course** — https://geometricdeeplearning.com/lectures/ · proto-book: https://arxiv.org/abs/2104.13478 · Veličković GNN lectures: https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures
- **Molecular ML** — DeepChem (https://deepchem.io/tutorials/) · Therapeutics Data Commons (https://tdcommons.ai/) · SchNet (https://arxiv.org/abs/1706.08566) · DimeNet++ (https://arxiv.org/abs/2011.14115) · MolFormer (https://arxiv.org/abs/2106.09553) · equivariant nets, Cohen & Welling (https://arxiv.org/abs/1602.07576)

Week 28 — Protein structure + equivariance, then integration (~35 hrs)

- **AlphaFold 2** — AlQuraishi explainer (https://www.youtube.com/results?search_query=AlphaFold+2+explained+AlQuraishi) · paper (https://www.nature.com/articles/s41586-021-03819-2) · ESM/ESMFold · e3nn/EGNN
- **Start the integration project** (see below).

Week 29 — Integration project + mock interviews + portfolio (~35 hrs)

- **Integration project (portfolio centerpiece):** something touching all three domains — e.g., predict a molecular property with a hybrid quantum-classical model pretrained on a chemical foundation model, deployed as a monitored service.
- **Mock interviews, recorded and reviewed:** 60-min ML systems design ×several; 45-min segmentation deep-dive (defend loss/architecture choices); 45-min quantum (defend VQE/ADAPT/DMET to a skeptical chemist); 30-min behavioral (STAR, 4 stories prepped).
- **Read 1 paper/day** from your subfields; **write 2 short blog/LinkedIn posts** distilling something learned — writing for an audience exposes shaky understanding.

Track M — Mathematics (parallel, folded into phase hours)

Linear algebra is front-loaded into **Phase 1 Weeks 1–2**. The rest threads into the phase where it pays off, so you learn it in context rather than in the abstract:

- **Multivariable + matrix calculus** (Phase 1–3 weekday slots) — 3B1B Calculus (https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr) · Matrix Calculus for DL (https://explained.ai/matrix-calculus/) — *~20 hrs*
- **Probability + information theory** (Phase 1, feeding loss functions; Phase 3, feeding generative models) — Harvard Stat 110 (https://projects.iq.harvard.edu/stat110/youtube) or MIT 6.041 (https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/) · MacKay, *Information Theory* ch 1–6 (https://www.inference.org.uk/itila/book.html) — *~45 hrs*
- **Optimization** (Phase 2 variational methods; Phase 3 training) — Boyd EE364A (https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · book https://web.stanford.edu/~boyd/cvxbook/) · Ruder, gradient-descent overview (https://www.ruder.io/optimizing-gradient-descent/) — *~45 hrs*
- **Advanced linear algebra / matrix methods** (Phase 1–2) — MIT 18.065 (SVD, PCA, low-rank, gradient descent through a linear-algebra lens) — https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/ — *~25 hrs*
- **Group theory & symmetry** (Phase 2 quantum; Phase 5 geometric DL) — group theory for physicists (https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures) · Bronstein GDL for the Lie-group/equivariance angle (https://geometricdeeplearning.com/lectures/) — *~25 hrs*

Key math checkpoints (spread across phases): derive cross-entropy from KL; show NLL minimization = MLE; derive the ELBO; explain mutual information and where it appears in contrastive losses (InfoNCE).

Cross-cutting habits (every week, no exceptions)

1. **Paper-of-the-week** — one paper, deeply read, derivations redone on paper. Don't break the streak.
2. **Whiteboard Friday** — one concept whiteboarded without notes; photograph it. These become your interview-prep deck.
3. **Re-implementation Saturday** — one concept from scratch in <200 lines. Repo: `weekly-rebuild/wk01-linalg`, etc.
4. **Sunday writeup** — a 1-page note explaining the week's deepest concept to a hypothetical junior. Teaching is the senior-level forcing function.
5. **Senior-engineer "war chest"** — one Obsidian/Notion doc collecting derivations, paper one-liners, gotchas, anti-patterns. By end of Jan you'll have 100+ pages and won't panic in an interview again.

Resource index (quick lookup)

Math — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · 3B1B Calc https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr · MIT 18.06 https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/ · MIT 18.065 https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/ · Stat 110 https://projects.iq.harvard.edu/stat110/youtube · MIT 6.041 https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/ · Boyd https://www.youtube.com/playlist?list=PL3940DD956CDF0622 · https://web.stanford.edu/~boyd/cvxbook/ · MacKay https://www.inference.org.uk/itila/book.html · Matrix Cookbook https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf · Matrix Calculus https://explained.ai/matrix-calculus/ · Ruder https://www.ruder.io/optimizing-gradient-descent/ · GDL lectures https://geometricdeeplearning.com/lectures/

Medical imaging — MRI Q&A https://mriquestions.com/index.html · OHBM https://www.youtube.com/@ohbmonline/playlists · Callaghan https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMl3PlpdW9TMP · AI for Medical Diagnosis https://www.coursera.org/learn/ai-for-medical-diagnosis · DigitalSreeni https://www.youtube.com/@DigitalSreeni/playlists · code https://github.com/bnsreenu/python_for_microscopists · Conv arithmetic https://arxiv.org/abs/1603.07285 · https://github.com/vdumoulin/conv_arithmetic · MONAI tutorials https://github.com/Project-MONAI/tutorials · MONAI YouTube https://www.youtube.com/@projectmonai/videos · nnU-Net https://github.com/MIC-DKFZ/nnUNet · TorchIO https://torchio.readthedocs.io/ · Stanford AIMI https://aimi.stanford.edu/education/educational-resources · CS231n https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv · https://cs231n.github.io/ · Metrics Reloaded https://www.nature.com/articles/s41592-023-02151-z *Segmentation papers:* U-Net https://arxiv.org/abs/1505.04597 · V-Net https://arxiv.org/abs/1606.04797 · Attention U-Net https://arxiv.org/abs/1804.03999 · nnU-Net https://www.nature.com/articles/s41592-020-01008-z · Loss survey https://arxiv.org/abs/2006.14822 · Generalized Dice https://arxiv.org/abs/1707.03237 · Focal Tversky https://arxiv.org/abs/1810.07842 · Boundary loss https://arxiv.org/abs/1812.07032 · ViT https://arxiv.org/abs/2010.11929 · TransUNet https://arxiv.org/abs/2102.04306 · Swin-UNet https://arxiv.org/abs/2105.05537 · UNETR https://arxiv.org/abs/2103.10504 · Swin UNETR https://arxiv.org/abs/2201.01266 · SAM https://arxiv.org/abs/2304.02643 · MedSAM https://www.nature.com/articles/s41467-024-44824-z · MedNeXt https://arxiv.org/abs/2303.09975

Deep learning core — Karpathy https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ · hub https://karpathy.ai/zero-to-hero.html · CS25 https://web.stanford.edu/class/cs25/ · https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM · CS336 https://stanford-cs336.github.io/spring2024/ · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_ · CS236 https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8 · Lilian Weng https://lilianweng.github.io/ · Annotated Transformer http://nlp.seas.harvard.edu/annotated-transformer/ · HF NLP https://huggingface.co/learn/nlp-course · HF CV https://huggingface.co/learn/computer-vision-course · HF Diffusion https://huggingface.co/learn/diffusion-course *Papers:* RoPE https://arxiv.org/abs/2104.09864 · RMSNorm https://arxiv.org/abs/1910.07467 · FlashAttention https://arxiv.org/abs/2205.14135 · GQA https://arxiv.org/abs/2305.13245 · SwiGLU https://arxiv.org/abs/2002.05202 · Switch/MoE https://arxiv.org/abs/2101.03961 · ConvNeXt https://arxiv.org/abs/2201.03545 · DINOv2 https://arxiv.org/abs/2304.07193 · MAE https://arxiv.org/abs/2111.06377 · DDPM https://arxiv.org/abs/2006.11239 · Score-based (blog) https://yang-song.net/blog/2021/score/ · Diffusion (blog) https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Quantum — IBM Quantum Learning https://learning.quantum.ibm.com/ · Watrous YouTube https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR · Basics https://learning.quantum.ibm.com/course/basics-of-quantum-information · Algorithms https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms · General Formulation https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information · Error Correction https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction · Qiskit YouTube https://www.youtube.com/@qiskit/playlists · QDA blog https://www.ibm.com/quantum/blog/iql-migration · PennyLane codebook https://pennylane.ai/codebook · PennyLane QML topic https://pennylane.ai/topics/quantum-machine-learning · PennyLane chemistry https://pennylane.ai/qml/demos_quantum-chemistry · MIT 8.04 https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/ · Quantum Country https://quantum.country/qcvc · Schuller https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic · Péré https://github.com/Christophe-pere/QML-Course · Hjorth-Jensen https://github.com/CompPhysics/QuantumComputingMachineLearning *Papers:* VQE https://www.nature.com/articles/ncomms5213 · ADAPT-VQE https://www.nature.com/articles/s41467-019-10988-2 · DMET-VQE https://arxiv.org/abs/2108.08987 · QAOA https://arxiv.org/abs/1411.4028 · Hadfield https://arxiv.org/abs/1709.03489 · QML=kernels https://arxiv.org/abs/2101.11020 · Barren plateaus https://www.nature.com/articles/s41467-018-07090-4 · Expressibility https://arxiv.org/abs/1905.10876 · Advantage-goal https://arxiv.org/abs/2203.01340 · PQC encodings https://arxiv.org/abs/2008.08605 · Pauli grouping https://arxiv.org/abs/1907.13117

Production / MLOps — CS329S https://stanford-cs329s.github.io/ · Made With ML https://madewithml.com/ · Full Stack DL https://fullstackdeeplearning.com/course/2022/ · Chip Huyen https://huyenchip.com/ml-interviews-book/ · Hamel Husain https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/ · Eugene Yan patterns http s://eugeneyan.com/writing/llm-patterns/ · Eugene Yan evals https://eugeneyan.com/writing/llm-evaluators/ · Anthropic contextual retrieval https://www.anthropic.com/news/contextual-retrieval · Anthropic Constitutional AI https://www.anthropic.com/news/constitutional-ai · LlamaIndex https://docs.llamaindex.ai/ · RAGAS https://docs.ragas.io/ · DSPy https://dspy.ai/ · Raschka https://magazine.sebastianraschka.com/ · HF multi-GPU https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM https://arxiv.org/abs/2309.06180 · ZeRO https://arxiv.org/abs/1910.02054

Drug discovery / Geometric DL — Bronstein GDL https://geometricdeeplearning.com/lectures/ · proto-book https://arxiv.org/abs/2104.13478 · DeepChem https://deepchem.io/tutorials/ · TDC https://tdcommons.ai/ · SchNet https://arxiv.org/abs/1706.08566 · DimeNet++ https://arxiv.org/abs/2011.14115 · MolFormer https://arxiv.org/abs/2106.09553 · Equivariant nets https://arxiv.org/abs/1602.07576 · AlphaFold 2 https://www.nature.com/articles/s41586-021-03819-2

A note on the discomfort you described

The feeling of going backwards — re-deriving things you already *use* — is the correct feeling for this transition, and it's slow and frustrating for the first 6–8 weeks. It isn't backwards. By the time you finish Phase 1's math front-load you'll read segmentation papers differently; by the end of Phase 2 you'll defend your own quantum work to a skeptical chemist without flinching. The hour budget is deliberately flexible so you can breathe on hard work-weeks. Trust the sequence, protect your rest, and adjust as you go — this is a scaffold, not a prison.